JANUARY 26, 2001

Inorganic Cume

Structural Determination and Structural Data

A. In a conventional small-molecule (12-100 atom) molecular structure determination using single crystal X-ray diffraction:

1. What is a common (desirable) ratio for the number of reflections to the number of parameters being refined in the structure?

2. Some atoms are reported as being refined "isotropically', some as "anisotropically". Explain.

3. What is the definition of the always-quoted "R-factor"? What is a "good" value for R?

4. In general terms, what is meant by "space group"?

5. What are the typical uncertainties (standard deviations, sigmas) for C-C bond angles and C-C-C bond angles in a "good" structure determination?

6. What are the new developments in X-ray diffraction instrumentation?

7. Explain, in general terms, the "phase problem" of X-ray crystallography.

8. How is it possible that the structure determination experiment can start with intensity data that is only reproducible within a few percent and yet the experiment concludes by quoting bond distances and angles with uncertainties of tenths of a percent?

B. Other Methods:

9. Briefly describe one of the methods which can be used to give bond distances and angles for a molecule in the gaseous state.

10. What are the significant differences between small molecule crystallography and protein crystallography?

11. What are the advantages of neutron diffraction? The disadvantages?

12. What are the advantages of synchrotron sources for diffraction studies?